Moreno Kristovan



Founding Engineer & Systems Architect | Technical Product Lead

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EXECUTIVE PROFILE

Systems-focused technical founder and software architect with deep execution in Apple platform architectures (macOS, iOS, watchOS), developer utilities, and mathematical simulation modeling. Proven track record leading multi-disciplinary engineering pods of up to 6 developers at the Apple Developer Academy, delivering production-grade applications under disciplined 2-week sprint constraints. Core technical competence spans creative workflow tools, real-time kinematic simulation engines, native platform integrations, and edge-efficient spatial compute. Combines grounded engineering execution with acute product empathy, architecture decision governance, and cross-border team coordination.

CORE TECHNICAL COMPETENCIES

Systems & Software Architecture

Swift 6 (Strict Concurrency), modular component contracts (<300 LOC), IPC, CoreGraphics, high-frequency IMU sensor fusion, client-server synchronization, macOS/iOS platform bridges.

Apple Platform Architecture

SwiftUI, AppKit integration, SwiftData, CoreHaptics, AudioToolbox, CoreMotion, CoreLocation (L1/L5 GNSS), TestFlight CI/CD release pipelines, Apple HIG compliance.

Mathematical Modeling & Telemetry

Kinematic approximations (friction, spring elasticity, Coulomb's Law), numerical integration, 2nd-order Butterworth digital signal filtering, continuous 6D rotation formulations (SO(3) projection).

Engineering Leadership & Governance

Architecture Decision Registers (ADRs), interface code contracts, sprint orchestration, technical mentoring, and cross-border engineering-design workflows.

SHIPPED PRODUCTS & CORE ENGINEERING PROJECTS

Penpal — Dedicated iPad-to-Mac Stylus Digitizer

July 2026 – Present

Product Architecture & Prototype | macOS, iPadOS, Swift, CoreGraphics

- Identified a core workflow friction for digital artists: professional creative software (Blender, Photoshop, Krita) runs natively on macOS where precision pen tablets are standard, yet many creators own an iPad with an Apple Pencil but cannot justify buying a separate expensive graphics tablet.
- Addressed the limitations of existing wireless second-screen solutions (compression lag, Wi-Fi latency, high battery drain, and lack of dedicated stylus fidelity) by exploring a direct wired digitizer pipeline that treats the iPad purely as a dedicated drawing tablet rather than a second monitor.
- Prototyped a low-overhead input capture pipeline harvesting high-fidelity Apple Pencil touches (pressure and tilt dynamics) over a stable wired connection, routing stylus events directly into macOS creative suites without the performance penalty of video streaming.

Reefora — Tactile Marine Ecological Simulation

July 2026 – August 2026

Technical Product Lead & Systems Architect | iOS, Swift 6, SwiftUI, SwiftData

- Directed end-to-end product architecture and delivery as Technical PM for an interactive marine conservation simulation built with restoration data from Living Seas Bali, leading a multicultural 5-developer squad across Uzbek, Kenyan, Indian, Australian, and Indonesian backgrounds.
- Architected a pure, decoupled domain `EcoEngine` executing mathematical models of Shannon species diversity, herbivore recruitment, and non-linear coral growth dynamics.
- Built natively in Swift 6 from Day 1 with complete strict concurrency (`-strict-concurrency=complete`) and zero data races, establishing strict component modularity under 300 LOC per file across the international engineering squad.
- Established an Architecture Decision Register (38 sequential ADRs) to maintain technical contracts and ship a feature-complete build within a 2-week MVP sprint.

BetterGitGraph — Developer Spatial Cognition Visualizer

August 2026 – September 2026

Creator & Lead Developer | TypeScript, VS Code API

- Designed and built an open-source VS Code extension rendering Git commit histories as a deterministic Directed Acyclic Graph (DAG) with visual branch routing to reduce mental cognitive load during complex branch operations.
- Implemented a two-stage verification state machine pairing destructive operations (rebase, reset, cherry-pick) with plain-language operational summaries and dry-run previews to prevent accidental code loss.
- Shipped to the VS Code Marketplace, organically reaching 100+ active developer downloads within seven days of release.

StereoPose 6D — Dual-View Spatial Estimation System

Dec 2024 – Sept 2025

Lead AI Researcher | Python, PyTorch, ConvNeXt V2 Nano

- Conducted undergraduate thesis research on edge-efficient 6D object pose tracking without relying on specialized LiDAR or depth sensor hardware.
- Eliminated gimbal lock and quaternion antipodal gradient discontinuities by implementing a continuous 6D rotation formulation (SO(3) projection via Gram-Schmidt orthogonalization).
- Evaluated Baseline, Siamese Late-Fusion, and 6-Channel Early-Fusion topologies using a lightweight ConvNeXt V2 Nano backbone, validated against ArUco-annotated geometric datasets.

NotSoDumbbells & Hapticle — Real-Time Telemetry & Tactile Dynamics

June 2026 – July 2026

Technical Product Lead | watchOS, iOS, Swift 6, CoreHaptics, AudioToolbox

- Built a deterministic 2nd-order Butterworth filter executing on Apple Watch S9/S10 SiP in under 0.02 ms, rejecting heavy ML runtimes to conserve battery while classifying acceleration profiles, rep tempo, and velocity metrics.
- Modeled physical forces (spring tension, rotary friction) into real-time state managers, binding force vectors to dynamic transient patterns in CoreHaptics and frequency-modulated AudioToolbox signals.

PROFESSIONAL & LEADERSHIP EXPERIENCE

Technical Product Lead & Systems Architect

March 2026 – Present

Apple Developer Academy @ BINUS Bali | Bali, Indonesia

- Directed technical architecture, sprint execution, and code review across multi-disciplinary engineering pods of up to 6 developers building native Apple platform applications.
- Enforced Swift 6 strict concurrency standards, automated TestFlight build delivery, and interface code contracts that eliminated cross-team integration blockers.
- Managed multicultural engineering pods (including talent across Central Asia, East Africa, South Asia, Australia, and Indonesia), aligning diverse international working styles around unified Architecture Decision Registers and native SwiftUI/AppKit component contracts.

Computer Science & Systems Instructor

May 2023 – December 2025

Timedoor Academy | Jakarta, Indonesia

- Instructed over 150 students across computer science fundamentals, algorithmic problem-solving, object-oriented design, and Python-based machine learning workflows.
- Mentored classrooms spanning diverse technical backgrounds (from secondary school students to university computer science undergraduates), developing strong technical translation and communication skills.
- Acted as lead technical unblocker across concurrent student projects, diagnosing memory leaks, logical deadlocks, and architecture flaws in real-time.

EDUCATION

Bachelor of Computer Science (S.Kom)

2021 – 2025

Universitas Bunda Mulia | Jakarta, Indonesia (GPA: 3.33 / 4.00)

- **Core Coursework:** Real-Time Systems, Computer Vision, Machine Learning Architecture, Distributed Systems, Data Structures & Algorithms.
- **Undergraduate Thesis:** *Comparative Analysis of Stereo Fusion Architectures for Real-Time 6D Pose Estimation Using Continuous Rotation Representations.*

TECHNICAL TOOLCHAIN & LANGUAGES

Languages: Swift 6, C, TypeScript, Python, Shell/Bash, SQL.

Apple Platforms & Frameworks: macOS, iOS, iPadOS, watchOS, AppKit, SwiftUI, SwiftData, CoreGraphics, CoreHaptics, CoreMotion, AudioToolbox, TestFlight.

Developer Infrastructure: Git, VS Code Extension API, Docker, Linux, Proxmox, GitHub Actions CI/CD.

Languages Spoken: English (Native), Indonesian (Native).